

Developing an Action Plan for Artificial Intelligence (AI) Innovation: A Framework Built on Transparency and Public Trust in AI

Darby Orcutt, Thomas A. Birkland, Veljko Dubljević, Kevin Lee, George List, William Rand, Benjamin Reading, and Munindar P. Singh



EXECUTIVE SUMMARY

As academic researchers engaged in Artificial Intelligence development and innovation, we share the vision of maintaining U.S. leadership in AI. Our universities (NC State University and NC Central University), two of the “world-class research institutions” noted in President Trump’s Executive Order 14179, exemplify this commitment. Our shared interest in promoting successful AI innovation led us to collaborate as the Center for AI in Society and Ethics (go.ncsu.edu/ai-society). As the Executive Order asserts, U.S. global leadership in AI requires “the right Government policies,” including robust funding for both basic and applied AI research, with a focus on interdisciplinary approaches such as ours - that foreground the social and human implications of AI.

AI is inherently shaped by and, in turn, shapes social institutions. AI technologies will empower new heights of human progress. The rise of AI will usher in a new industrial revolution, characterized by massive shifts toward human-machine collaboration, transforming the human work and the workforce. It will boost medical care and scientific discoveries into overdrive, rapidly impacting the quantity and quality of people’s lives. AI will utterly transform the core dynamics of economies, of education, and of the very ways that humans interact with and relate to one another. The positive impacts that AI innovation could have on people and society, both in the near and short term, cannot be overstated.

Yet, public apprehension and wariness of AI, evidenced in surveys of public attitudes, underscores the need for transparency and human-centric AI development. As Pflanzner et al. (2023) note, “AI does not simply, automatically, and seamlessly integrate into our daily lives and social institutions. Rather, it directly reshapes social, cultural, and economic structures and affects the lives of individual citizens in profound and often tacit, unpredictable, or morally questionable ways.” This highlights the potential for AI to reweave or even tear our socioeconomic fabric, impacting not just productivity and safety but also autonomy and dignity. AI innovation must be appropriately carried out and governed to ensure that its massive potential benefits indeed manifest. Governance principles that prioritize transparency and consent—fundamental to democracy—must guide AI innovation.

Existing frameworks for technology oversight are often reactive, slow to adapt, and lack adequate public input. Unlike past technologies, AI pervades every sector and increasingly operates autonomously, requiring a proactive and holistic approach to governance. Public concerns about AI undermine its adoption, consumer confidence, and support for its development. To address these challenges and facilitate effective and fair human-AI teaming (see: Pflanzner et al. 2022), AI governance must integrate both technical and public expertise throughout the development cycle.

Unsuitability of Current Governance Frameworks

Whereas the U.S. has instituted various governance mechanisms across multiple domains, including the SEC for securities, NHTSA for highway safety, the FCC for communications, and Institutional Review Boards (IRBs) for research involving humans, these frameworks are often ill-suited for AI governance. Current mechanisms are generally reactive, slow to adapt, and lack support for ongoing public engagement. In the case of AI, the stakes are much higher, with concerns ranging from bias and discrimination to the catastrophic risks posed by mis-deployed AI systems that could endanger human lives. These existing frameworks, built in an era of slower technological change, are not equipped to deal with the rapid evolution of AI technologies.

Moreover, current governance mechanisms often operate in silos, with separate agencies handling different technology domains. This fragmented approach is problematic for AI, which, by its nature, will pervade all sectors of society and the economy. Unlike previous technologies, AI has the capacity to operate autonomously, making decisions without human oversight and shifting responsibility in unprecedented ways. Therefore, a flexible, standards-based governance framework is necessary—one that provides overarching guidance across sectors while addressing the unique challenges posed by AI applications.

AI Innovation Requires a Foundation of Transparency and Consent

Successful AI innovation incorporates both technical and public expertise to ensure AI development is innovative, aligned with the public good, and appropriately mitigates risks. To the greatest extent technologically feasible, AI decision-making must be explainable, accountable, and subject to public oversight. While open source approaches should be encouraged, a balance of interests in terms of protecting intellectual property and maintaining competitive advantage should be recognized as legitimate. These principles align with similar administrative law doctrines, consumer protection laws, and financial disclosure mandates, such as the Equal Credit Opportunity Act.

Specifically, standards of governance should:

- **For Transparency:**
 - Require AI systems to include self-monitoring capabilities that track performance on both domain-specific and domain-independent key performance indicators (KPIs).
 - Ensure that AI systems report their performance transparently to relevant stakeholders.
 - Establish pathways for continual improvement and correction, not just for individual decisions but for the overall operation of AI technologies. This is essential as stakeholders may discover new needs, imperfections, or unintended consequences after interacting with the system.
- **For the Public's Consent:**
 - Ensure that AI systems are auto-correcting so mistakes can be undone before they entrench undesirable practices.
 - Implement standards for determining the appropriate level of scrutiny for a particular AI technology based on its domain of application, with the potential for revisiting these standards as technologies evolve.
 - Actively engage affected communities and other stakeholders in defining their needs, values, and risk attitudes. Public consultation should be integral to the governance of AI to maintain trust and transparency.

Given the intimate nature of many AI applications, governance frameworks must ensure that the perspectives of affected communities are included in decision-making. AI systems should be treated as ongoing research projects, where interdisciplinary teams, including relevant human-centric experts (for example, in psychology, economics, and ethics) are involved in every stage of development.

A Proposed AI Governance Framework

Built upon this foundation of transparency and consent, we propose flexible standards of governance for AI (rather than sheer regulation) to ensure effective, human-centered outcomes while fostering innovation. This framework values oversight, adaptability, and compliance with the U.S. Constitution and existing regulations and statutes.

Accordingly, our proposed governance framework consists of five pillars:

1. **Continual Monitoring and Improvement:** AI systems should support continual self-monitoring, track performance against defined benchmarks, and facilitate ongoing updates. These principles are informed by similar policies in FDA medical device regulations, algorithmic trading oversight, and real-time safety monitoring in self-driving vehicle laws.

2. **Bias Prevention:** The risk of systemic bias in AI applications requires safeguards. This governance framework mandates fairness audits, independent oversight, and compliance with equal opportunity and civil rights statutes. It aligns with the Equal Employment Opportunity Commission's (EEOC) disparate impact doctrine and recent Department of Justice guidance on AI-driven hiring discrimination.
3. **Risk-Proportional Scrutiny:** AI governance must adapt to the risk level of applications. High-risk applications, such as those in healthcare and criminal justice, should undergo heightened scrutiny, while lower-risk applications may require lighter oversight. This approach mirrors the FDA's risk-tiered classification of medical devices and IRB oversight models. Risk is understood to be a function both of the likelihood of an undesirable outcome or incident and the magnitude of the harm that would result from that incident. Like with the FDA and with the IRB's oversight of research, the greatest attention must be paid to those AI applications that pose the greatest risk. Even outcomes that are unlikely require considerable attention if the consequences of an error are sufficiently grave. These risks must be carefully weighed by competent authorities in the public and private sector, and systems must be designed to ensure appropriate monitoring and oversight.
4. **Active Stakeholder Engagement:** AI governance must incorporate multiple perspectives, including technologists, ethicists, policymakers, and the general public. This approach is supported by participatory rulemaking processes, such as The Federal Advisory Committee Act (FACA), reinforcing the necessity of inclusive governance.
5. **Support for Interdisciplinary AI Research:** AI innovation requires not only funding, but funding of basic and integrative research encompassing all aspects of AI, from computer science to neurocomputational ethics to domain-specific applications, and more. In addition, researchers require access and clear guidelines in using proprietary data crucial to AI training. The Fair Use Doctrine was set forth in Section 107 of the U.S. Copyright Act to enable research in the pre-digital age; it must be made clear that this still applies in the digital world as a foundational Doctrine protecting research uses from obstruction by terms of use, licensing, or other agreements. Generative AI innovation cannot happen without content, and actual or feared litigation in this area is already stifling U.S. competitiveness in AI.

This governance framework finds support in existing legislative directives, including the **National Artificial Intelligence Initiative Act of 2020 (NAIIA)** (15 U.S.C. 9401 et seq.), which establishes interagency coordination and defines national AI priorities, and aligns with the **Federal Trade Commission (FTC) Act** (15 U.S.C. 41 et seq.), which provides statutory authority for enforcement against deceptive AI practices.

Conclusion

AI governance should not merely react to failures but should proactively shape the responsible development and deployment of AI technologies. By adopting a rational and ethical sociotechnical perspective (see: Chopra & Singh 2018; Dubljević et al. 2021) and leveraging AI's capabilities, we can create an effective governance model that fosters innovation while safeguarding public trust and societal well-being. Standards of transparency and informed consent, together with interdisciplinary engagement, will ensure that AI technologies are developed with the public's best interests in mind and are capable of serving the common good.

References:

- Chopra, A. K. & Singh, M.P. (2018): Sociotechnical systems and ethics in the large. In Proceedings of the AAAI/ ACM Conference on Artificial Intelligence, Ethics, and Society (AIES), pp. 48–53, New Orleans, ACM. DOI: 10.1145/3278721.3278740.
- Dubljević, V., List, G., Milojevich, J., Ajmeri, N., Bauer, W.A., Singh M.P., Bardaka, E., Birkland, T.A., Edwards, C.H.W., Mayer, R.C., Muntean, I., Powers, T.M., Rakha, H.A., Ricks, V.A. & Samandar, M.S. (2021): Toward a Rational and Ethical Sociotechnical System of Autonomous Vehicles: A Novel Application of Multi-Criteria Decision Analysis. PLoS ONE, DOI: 10.1371/journal.pone.0256224.
- Pflanzer, M., Dubljević, V., Bauer, W.A., Orcutt, D., List, G.F. and Singh, M.P. (2023): Embedding AI in Society: Ethics, Policy, Governance and Impacts (Editorial), *AI & Society* 38:1267-1271. DOI: 10.1007/s00146-023-01704-2.
- Pflanzer, M., Traylor, Z., Lyons, J., Dubljević, V. & Nam, C.S. (2022): Ethics of Human-AI Teaming: Principles and Perspectives. *AI and Ethics*, DOI: 10.1007/s43681-022-00214-z.

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.